

tradeforme — Next Steps

Action list of things only **you** can do. Everything below is blocked on your input, your credentials, or your decision — not on more code.

Where the project stands

Repository	https://github.com/EtixP/tradeforme (private)
Branch	main
Milestones done	M1, M2, M3 (deterministic extraction, 96.6% OK), M4, M5 (signals stored, v2 with KOSP)
Tests passing	92 / 92
Disclosures in DB	514,304 across 488 trading days (Jun 25 2024 – Jun 24 2026, full 24 months)
Supply-contract events	3,531 candidates → 3,412 OK + 119 manual-review (96.6% / 3.4%), 0 blocked
Real contract values extracted	3,412 events with verified contract_value, prior_year_revenue, ratio
Candidate signals stored	693 (v2 strategy: ratio ≥ 0.08, KOSPI only, skip government counterparty). Mean strength
Price data source	pykrx (free, no auth)
LLM extraction	Code complete (prompts + validator + client) — optional now that deterministic parser wo
Broker integration	PaperBroker (historical fills) ready; KIS broker needs your credentials

Threshold sweep on 24 months / 3,412 events

Net of 0.313% roundtrip cost (commission + VAT + sale tax + slippage). Loop-3 ran the same sweep on only 514 events (3 months); the comparison column shows how those earlier numbers held up after 6.6× more data.

Threshold	n (24mo)	T+1 net	T+5 net	T+5 win%	PF (5d)	Median 5d	Loop-3 T+5 (n=514)
0.00 (all)	3,412	−0.12%	+0.35%	44.1%	1.21	−0.68%	+1.21% (n=514)
0.08 «	2,176	−0.07%	+0.22%	42.9%	1.16	−0.73%	+0.55% (n=333)
0.15	1,177	−0.16%	+0.56%	44.0%	1.25	−0.71%	+0.89% (n=169)
0.20	819	−0.06%	+0.59%	42.9%	1.25	−1.07%	+1.15% (n=110)
0.30	470	+0.18%	+0.50%	41.2%	1.21	−1.29%	+3.10% (n=59) «
0.50	217	+0.47%	+0.92%	38.1%	1.31	−1.79%	+2.67% (n=23)
1.00	66	−0.06%	−1.83%	32.3%	0.69	−3.98%	—

The big finding: the ≥ 0.30 "edge" from Loop 3 was largely noise. Going from 59 to 470 events, the T+5 net mean dropped from +3.10% to +0.50% and profit factor from 1.93 to 1.21. Median is negative at every threshold. The 1.00+ row is actually negative — suggests very large contracts may be a slight negative

signal (possibly representing distressed wins or unusual situations).

Honest conclusion: after 24 months and 3,412 events, the major-supply-contract event does NOT produce a clean tradable post-disclosure drift after costs. Mean net returns at every threshold are within noise of zero, win rates are 38–46%, medians are negative. This is exactly the kind of "strong negative result" CLAUDE.md anticipates — it answers the original research question. The market reacts efficiently to these disclosures by event-day close.

What can still rescue this: (a) sub-segment by counterparty type, sector, market cap, or recent volatility — maybe certain slices DO work. (b) try other event types (buybacks, dilutive financing, halt/resumption). (c) different exit horizons (T+10, T+20). (d) the M5b ML filter on richer features. Each is a real hypothesis to test — none is guaranteed to work.

Loop-6 subgroup + walk-forward results

Sliced the 3,377 events by market, ratio bucket, contract value, day-of-week, calendar quarter, and counterparty type (regex heuristic). Then walk-forward validated the most-positive subgroup and the most-negative subgroup across four 6-month windows. Honest reading below.

Subgroup	Aggregate T+5	Aggregate PF	Walk-fwd verdict (4 windows)
KOSPI / ratio 0.15-0.30	+1.59% (n=199)	1.59	ALTERNATES (+/-/-/+) — not robust
KOSPI (any ratio)	+0.63% (n=1708)	1.22	3/4 windows positive — modest but holds
KOSDAQ (any ratio)	+0.06% (n=1669)	1.02	no edge
Government counterparty	−2.14% (n=90)	0.51	3/4 windows NEGATIVE (sometimes −5%)
Other_korean_corp	−0.65% (n=417)	0.84	consistently negative
Ratio 1.00+	−1.83% (n=66)	0.64	small n; likely distressed/unusual events

Two actionable findings:

- **Government-counterparty contracts consistently underperform.** 3/4 walk-forward windows show net T+5 returns of –5% to –0.5%. Aligns with intuition: government contracts are pre-priced, low-margin, expected. We can't short (CLAUDE.md rule), but this is a **strong negative filter** — the strategy should skip long signals where counterparty is government.
- **KOSPI > KOSDAQ** in 3/4 windows. Aligns with the liquidity-premium hypothesis. Worth restricting the strategy to KOSPI-only.

Two findings ruled out: the KOSPI/0.15-0.30 subgroup that looked best in aggregate (+1.59% T+5) alternates positive/negative across windows — a classic small-sample fluke. The 0.30+ ratio "edge" from Loop 3 stays dead at scale. Median return remains negative at every threshold.

Raw CSV: data/subgroup_analysis.csv (3,377 events x 6+ feature columns).

Raw CSV: data/event_study_results.csv (3,531 events, all horizons, gross). Loop-3 log of the smaller dataset is preserved in git history.

Action items

Sorted by priority. P1 unblocks the next milestone. P2 hardens the existing system. P3 is optional polish.

Priority	What	Why	How
P1	Decide on the strategic pivot	Loop9 paper highlights loop9 the v2 strategy which is path 42/next leaked (a) is a list of descriptive	
P1	Get an LLM API key (Anthropic/Ms/OpenAI)	Needed for ML initiative cost clear (trooper/carry type/language strength) Addressed	
P2	Rotate the DART API key	The current key briefly appeared in chatgpt during the 2023 smoke test (2023) Replace xDg	
P2	Set up GitHub branch protection	Once live trading code exists, you don't want a stray PR to make it to prod	Set up GitHub branch protection
P2	Back up data/kdtb.db	111k rows of ingested DART data	Easiest: rent a 10GB S3 bucket, backup 5 min daily, keep 90 days
P2	Schedule daily DART ingestion	Right now ingestion only runs when you invoke the CLI manually	Use cron or systemd timer to run CLI daily
P2	Obtain KIS broker credentials	Required for Milestone 6 (broker integration)	Go to KIS website, register, get API key
P2	Decide whether to raise min_confidence threshold	Current threshold sweep requires fine-tuning of model	Test different thresholds, see if model performance improves
P2	Review risk thresholds in config	Default: 100k with CLAUDE.md (100k)	Open config, adjust risk thresholds, save
P3	Install gitleaks pre-commit hook	Catches an accidental commit of .env	brew install gitleaks & gitleaks pre-commit
P3	Decide if PDF action lists should be included	Currently only STEPS.pdf will be included	Decide if PDFs should be included in the package
P3	Decide on Korean market holidays	Big trading the ingestion CLI runs even on holidays	Optional: add holiday calendar (only for Korea)

Where to find things

Project root	/Users/jsp2022310/Desktop/tradeforme
Repo on GitHub	https://github.com/EtixP/tradeforme
Project spec	CLAUDE.md (in repo root)
Source code	src/kdtb/ — schemas, config, data, strategy, risk, storage
Tests	tests/ — run with: <code>.venv/bin/pytest</code>
DART ingestion CLI	<code>scripts/ingest_disclosures.py --date YYYY-MM-DD</code>
Event study CLI	<code>scripts/run_event_study.py [--limit N]</code>
Parse docs CLI	<code>scripts/parse_supply_contracts.py [--limit N] [--skip-existing]</code>
Filtered event study	<code>scripts/run_filtered_event_study.py [--min-ratio 0.08]</code>
Strategy runner	<code>scripts/run_strategy.py [--min-ratio 0.30] [--dry-run]</code>
DB backup	<code>./scripts/backup_db.sh</code> — gzipped SQL dump under <code>data/backups/</code>
Walk-forward	<code>scripts/walk_forward.py</code> — reproduces the v0/v1/v2 4-window table
Paper backtest (NEW)	<code>scripts/run_paper_backtest.py</code> — realized-PnL with 3 execution assumptions
Local DB (gitignored)	<code>data/kdtb.db</code> — SQLite, inspect with <code>sqlite3</code> or DB Browser
Event study CSV	<code>data/event_study_results.csv</code>
Secrets (gitignored)	<code>.env</code> — <code>DART_API_KEY</code> lives here
Config	<code>config/default.yaml</code> , <code>config/paper.yaml</code> , <code>config/live_tiny.yaml</code>
Live-trading guard	<code>src/kdtb/config.py</code> — raises if YAML says LIVE without <code>ENABLE_LIVE_TRADING=true</code> in

How to verify the project is healthy

Run these from the repo root any time you want to confirm nothing is broken:

```
.venv/bin/pytest
```

```
.venv/bin/python -m kdtb.main
```

```
.venv/bin/python scripts/ingest_disclosures.py --date $(date +%Y-%m-%d)
```

What Claude built across the autonomous loops

Loop 1 (M4 + M5):

- **data/market_data_client.py** — pykrx wrapper for OHLCV + event-time returns
- **scripts/run_event_study.py** — batched event study CLI
- **strategy/major_supply_contract.py** — M5 strategy engine (was placeholder)

Loop 2 (cost model + M3 prep + paper broker):

- **backtest/cost_model.py** — Korean equity costs
- **backtest/metrics.py** — per-trade metrics
- **interpretation/llm_prompts.py**, **extraction_validator.py**, **llm_client.py**
- **broker/base.py** + **broker/paper** **broker.py**

Loop 3 (deterministic extraction + filtered event study):

- **data/dart_client.py** — new fetch_document_text() + ZIP/HTML/UTF-8 extractor
- **interpretation/deterministic_parser.py** — regex parser for ■■■■■■■■■■■■ (voluntary + mandatory variants)
- **storage/extraction_store.py, scripts/parse_supply_contracts.py, scripts/run_filtered_event_study.py**
- 16 new test cases

Loop 4 (end-to-end signal generation + parser refinement):

- Parser refinement: distinguishes 'value undisclosed' (literal dash) from 'missing entirely'. 0 blocked rows now.
- **storage/signal_store.py + scripts/run_strategy.py**
- 331 candidate signals stored in DB at default threshold (Loop 4 snapshot)
- 4 new test cases

Loop 7 (codified the walk-forward filters into the strategy):

- Added counterparty_name + counterparty_type fields to Extraction schema, with idempotent ALTER TABLE migration in init_db.
- Parser now classifies counterparty (government / large_corp_korean / other_korean_corp / foreign / unknown) using the same regex heuristic the subgroup analysis used.
- Backfilled counterparty_type for all 3,531 existing extractions (94 government, 154 foreign, 331 large_corp_korean, 439 other_korean_corp, 2513 unknown).
- Added kospi_only and excluded_counterparty_types params to MajorSupplyContractStrategy. Defaults in config: kospi_only: true, excluded_counterparty_types: ["government"].
- **Real PnL improvement** from the filters (still on the same 3,412 OK extractions):

Variant	n	T+1 net	T+5 net	T+5 win%	PF 5d	Notes
v0 — no filters	3,377	-0.12%	+0.35%	44.1%	1.10	baseline (Loop 5)
v1 — ratio ≥ 0.08	2,149	-0.07%	+0.22%	42.9%	1.06	CLAUDE.md default
v2 — KOSPI + skip-gov	684	+0.17%	+0.67%	46.6%	1.25	Loop-7 filters

The v2 strategy nearly tripled T+5 net mean over v1 (+0.67% vs +0.22%), pushed win rate from 42.9% to 46.6%, and lifted profit factor from 1.06 to 1.25. T+1 mean turned positive for the first time. Median is still -0.31% at T+5 — most individual trades still lose, but winners are bigger.

Honest caveats: 684 events over 24 months is ~28/month. Win rate is still <50%. PF 1.25 means you win \$1.25 for every \$1 lost — thin margin once execution slippage and the 18 blocked rows drift the numbers. This is "real progress", not "ready to trade".

- 92 tests still passing (+4 new for the filters).
- Tests cover: `kospi` only rejects KOSDAQ, `kospi` only off lets KOSDAQ through, `excluded` counterparty types rejects

matching, None counterparty_type still passes.

Loop 8 (walk-forward validation of v0/v1/v2):

The v2 aggregate looked promising; this loop verified that the improvement holds across 4 non-overlapping 6-month windows (the real test of "edge vs curve-fit").

Window	v0 T+5 / PF (n)	v1 T+5 / PF (n)	v2 T+5 / PF (n)
Jul24-Dec24	+0.04% / 1.01 (785)	−0.38% / 0.90 (498)	+1.57% / 1.58 (179) «
Jan25-Jun25	+0.59% / 1.23 (713)	+0.51% / 1.19 (463)	+1.11% / 1.65 (138) «
Jul25-Dec25	+0.41% / 1.15 (930)	+0.67% / 1.25 (573)	−0.55% / 0.79 (200)
Jan26-Jun26	+0.45% / 1.10 (906)	+0.15% / 1.03 (591)	+1.01% / 1.28 (163) «
Aggregate	+0.37% / 1.11 (3,334)	+0.25% / 1.07 (2,125)	+0.72% / 1.27 (680)
Positive windows	4/4	3/4	3/4

Verdict: v2 holds up. 3 of 4 windows positive, average +1.23% in the three winning windows, −0.55% in the losing one. The losers are smaller in magnitude than the winners. Aggregate T+5 net mean +0.72%, profit factor 1.27, win rate 47.1%. v1 is consistently the worst of the three — the ratio filter alone removes signal without adding edge. The KOSPI + skip-government conjunction is what's doing the work.

Still not a strategy ready for real money. 47% win rate, one losing window in four, PF 1.27 (■1.27 won per ■1 lost) — thin margin once execution slippage drifts the numbers. But it IS the first thing in this project that survives walk-forward validation. The next moves (paper broker simulation, then ML on top) are now standing on real ground.

Reproduce with: `.venv/bin/python scripts/walk_forward.py`

Loop 9 (realistic-execution backtest — the painful finding):

Re-priced the 684 v2 signals under three entry/exit assumptions. The "idealized" scenario matches the event study (buy event-day close, sell T+5 close), which assumes a fill we usually *can't* get as a retail trader.

Scenario	Mean	Median	Win%	PF	■/trade (on ■30k)	Total over 684 trades
idealized (t0 close → t+5)	+0.67%−0.31%		46.6%	1.25	+■201	+■137,779
realistic (t+1 close → t+5)	+0.14%−0.82%		42.0%	1.05	+■43	+■29,140
conservative (t+2 close → t+5)	+0.32%−0.31%		44.2%	1.15	+■97	+■66,078

The painful finding: under realistic execution (entering at the T+1 close instead of the event-day close), **most of v2's edge disappears.** Mean net per trade falls from +0.67% to +0.14%, PF from 1.25 to 1.05, win rate from 47% to 42%. Total realized PnL across 684 trades in 24 months would be roughly +■29,000 — ~■42/trade. Easily wiped out by execution slippage we haven't modeled.

Distribution shows why: 52% of v2 trades lose money under realistic execution (20% lose more than 5%). The 16% of trades that gain >5% just barely outweigh the losers. This is a strategy whose entire "edge" is concentrated in the event-day reaction. If you can't trade at the close, you don't get the edge.

What this means strategically: the major-supply-contract drift is real but largely captured by the time a retail trader can act. Three realistic paths forward: (a) build infrastructure to react WITHIN seconds of DART publication — that's how Korean HFT desks trade these. CLAUDE.md explicitly excludes this as a non-goal. (b) Test other event types that may have longer-lived reactions: buybacks, dilutive financing, halt/resumption. (c) Accept the verdict and document the negative result as the project's deliverable. The deterministic-pipeline + walk-forward methodology is itself valuable; the strategy choice was the wrong horse.

Reproduce: `.venv/bin/python scripts/run_paper_backtest.py`
Per-trade detail: `data/paper_backtest_v2.csv`

Loop 5 (24-month backfill + scale-out + the negative finding):

- CLAUDE.md updated: ML/LLM allowed for extraction + feature enrichment + filtering, but NOT in the order-placing decision. New **Milestone 5b** added (ML-enriched signal filter).
- **scripts/backup_db.sh** — gzipped SQL dumps to `data/backups/`, keeps last 7, ready for cron.
- **24-month DART backfill:** 111,622 → **514,304 disclosures** across 488 trading days (Jun 2024 – Jun 2026).
- **Re-ran deterministic parser:** 533 → **3,412 OK extractions** (96.6% success rate held up at scale, 0 blocked).
- **Re-ran event study + threshold sweep:** the 30%+ edge from Loop 3 (T+5 +3.10%, PF 1.93) collapsed to T+5 +0.50% / PF 1.21 on 8× more data. Median negative at every threshold.
- **Event study script now checkpoints CSV every 100 stocks** — one of yesterday's runs hung overnight; we no longer lose hours of fetches.

Suite: 28 → 43 → 68 → 84 → **88** passing tests. Zero regressions across five loops.

Honest caveats

- The naive event study uses title-only filtering. It does not separate small contracts from large ones, so noise dominates. M3 (LLM extraction) is needed to filter by contract/revenue ratio.
- The event date used is the DART receipt date (YYYYMMDD). Intraday timing of the disclosure is not captured. If a disclosure is filed at 14:00 KST, our 'T+1' return actually includes the price reaction from 14:00 to next-day close, not a clean next-day-only return.
- Returns are raw close-to-close, not benchmark-adjusted. A market-wide rally on event days would inflate the mean.
- pykrx scrapes the KRX website. It's free but can be rate-limited. We sleep 0.15s between calls.
- LIVE_TINY broker code is not yet implemented. Even with KIS credentials, no real orders can be placed yet.